What conditioning on sampled target moments costs a population-adjusted indirect comparison

A simulation study answering part of catalog problems MIS-03 and EST-07

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Abstract

Population-adjusted indirect comparisons match individual patient data from a source trial to a target trial known only through a published baseline table. Every matching-adjusted indirect comparison implementation we could examine treats the target's reported covariate means and standard deviations as exact constants, though they are sample estimates carrying their own sampling error, and no study has measured what that costs in anchored MAIC.

Registered result. The prespecified analysis is **uninformative**. Its gates require the reference interval and the negative controls to be nominal, and at poor covariate overlap they are not: the Wald sandwich undercovers by up to seven percentage points for every method compared, including scenarios containing no effect modification at all, where the population omission is exactly zero and the four intervals should very nearly coincide. That is a failure of the source variance estimator under extreme weighting, not a target-moment effect, and it invalidates the absolute-coverage analysis wherever it occurs.

Exploratory result. Restricting to the 196 of 252 scenarios where the reference and its matched control are nominal, conditioning on the reported moments omits between -9.5% and 14.9% of the variance and moves coverage of a nominal 95% interval between 92.1% and 96.2%. This restriction selects on an observed outcome and the results are labelled exploratory throughout.

An identity, not a finding. The net omitted component is, to first order, $(1-2\kappa)\mathrm{Var}_T\{\tau(X)\}/n_T$, where κ is how far the target trial's own treatment effect is modified by the same covariates as the source trial's. Its sign is therefore fixed analytically rather than discovered; regressing the measured omission on this prediction gives a slope of 1.003 (SE 0.042). What the simulation adds is the magnitude at realistic sample sizes, and the observation that κ is not identified from the inputs a matching-adjusted comparison ordinarily uses.

Practical implication, restricted. The two corrections an analyst could deploy add only the positive moment-variance term and omit the cross-covariance, so they are **partial**. Analytically they are closer to the correct variance than doing nothing only when $\kappa < 1/4$; between 1/4 and 1/2 they are worse than doing nothing, and the design contains no interior κ with which to test this. In the exploratory restricted set, at $\kappa \in \{0, 0.5, 1\}$ with **exactly** multivariate-normal covariates and a single equicorrelation discrepancy, they kept coverage inside 93% to 97% in every scenario, at a median interval-width cost of 2.24%. That figure applies under exact multivariate normality and may differ substantially under skewed, bounded, categorical or rounded covariates, which are untested. Extrema sat within about one Monte Carlo standard error of the band edges.

1 The problem

A population-adjusted indirect comparison estimates the relative effect of two treatments never compared in one trial. Individual patient data are available for a source trial comparing A with a common comparator C;

for the target trial comparing B with C, only a publication is available. Matching-adjusted indirect comparison (1) reweights the source participants so their covariate distribution matches the target's, transports the A versus C effect, and subtracts the target trial's own B versus C effect (2).

Everything hangs on a small table: per covariate, a mean and a standard deviation, with the sample size. Those numbers are the matching targets.

They are sample moments computed from n_T participants and carry sampling error of order $n_T^{-1/2}$. An interval that conditions on them omits that variability whenever the intended estimand refers to a target *superpopulation* rather than to the particular people who enrolled.

Catalog entry MIS-03 states the position. The weight-estimation half of the uncertainty problem has been benchmarked: (3) compared conventional, ESS-rescaled, robust sandwich and bootstrap variance estimators across 108 scenarios for binary and time-to-event outcomes, finding conventional estimators anticonservative under poor and moderate overlap and all methods valid under strong overlap. That is a specific set of settings, not a general guarantee, and section 5 shows the conventional sandwich failing outside them. The published-moment half is untouched, because a baseline table arrives without the covariance among its own entries.

Two results in the transportability literature propagate target-summary uncertainty for entropy-balancing estimators (4,5). Neither is framed for the anchored two-trial setting in which the target moments and the target treatment effect come from the same participants, and neither has been carried into indirect-comparison software. The general idea that estimated calibration totals contribute variance, and that overlapping samples induce covariance, is long established in survey calibration and two-phase sampling; nothing here claims that as new. What is new is the quantification for anchored MAIC and the identification of which term is missing.

2 The estimating system and what it omits

The source trial contributes n_S participants with covariates $X_i \in \mathbb{R}^3$, treatment indicator A_i and outcome Y_i. Write $h(X) = (X_1, X_2, X_3, X_1^2, X_2^2, X_3^2)^\top$; matching on h is matching on the reported means and standard deviations. A publication reports the unbiased sample standard deviation s_j, and the second raw sample moment is recovered as $\hat{m}_{2j} = \{(n_T - 1)/n_T\} s_j^2 + \bar{x}_j^2$, so the calibration target is exactly the sample mean of $h(X)$ over the target participants.

MAIC solves, over the source participants,

$$\sum_{i=1}^{n_S} w_i(\lambda)\,\{h(X_i) - \hat{m}_T\} = 0, \qquad w_i(\lambda) = \exp\{\lambda^\top h(X_i)\}, \tag{1}$$

the method-of-moments weighting of (1), the dual of entropy balancing (6). Stacking Equation 1 with the two weighted arm-mean equations gives an M-estimator (7) in $\psi = (\lambda, \mu_A, \mu_C)$ with score

$$u_i = \big(\, w_i\{h(X_i) - \hat{m}_T\},\ \ \mathbb{1}(A_i)\, w_i(Y_i - \mu_A),\ \ \mathbb{1}(C_i)\, w_i(Y_i - \mu_C)\,\big)^\top,$$

transported effect $\hat{\theta}_{AC} = \hat{\mu}_A - \hat{\mu}_C$, and anchored estimate $\hat{\theta}_{AB} = \hat{\theta}_{AC} - \hat{\theta}_{BC}$. With $\hat{A}$ the empirical Jacobian of the mean score, $\hat{B} = n_S^{-1} \sum_i u_i u_i^\top$ and $c = (0, \dots, 0, 1, -1)^\top$,

$$V_S = c^\top \hat{A}^{-1} \hat{B} \hat{A}^{-\top} c \,/\, n_S. \tag{2}$$

Equation 2 conditions on $\hat{m}_T$. Differentiating the estimating system with respect to the reported moments, and using $\partial u_{i,\lambda}/\partial \hat{m}_T^\top = -w_i I_6$,

$$J \;=\; \frac{\partial \hat{\theta}_{AC}}{\partial \hat{m}_T} \;=\; -\,c^{\top} \hat{A}^{-1} C, \qquad C = \begin{pmatrix} -\bar{w} I_6 \\ 0 \\ 0 \end{pmatrix}. \tag{3}$$

J is zero in the population when no covariate in h modifies the treatment effect, which is why the study's negative control has no effect modification. In a finite randomized source trial chance imbalance makes $\hat{J}$ nonzero, so the four intervals do not coincide exactly even there; section 5 reports how much.

Two consequences. If $\hat{m}_T$ were independent of everything else the interval would simply be too narrow, by $J^{\top}\Omega_{hh}J/n_T$. But in the ordinary case $\hat{m}_T$ and $\hat{\theta}_{BC}$ come from **the same people**. Writing ϕ_{BC} for the influence function of the target arm-mean difference, the unconditional variance uses the joint covariance of $\{h(X), \phi_{BC}\}$ with gradient $(J, -1)$:

$$V = V_S + (J, -1)^{\top}\, \Omega_T \,(J, -1)\,/\,n_T. \tag{4}$$

The cross term $-2\,J^{\top}\mathrm{Cov}\{h(X), \phi_{BC}\}/n_T$ is negative when the target treatment's effect is modified by the same covariates as the source treatment's. Conditioning therefore omits a positive term and a negative term whose balance depends on a quantity that appears nowhere in a publication.

2.1 An identity that fixes the sign

Under this study's mechanism the balance is exact to first order. With $\tau(X) = \tau_0 + s\,g(X)$ in the source and $\tau_0 + \kappa s\,g(X)$ in the target, and g linear in h with coefficient vector b, the population sensitivity is $J = sb$ and $\mathrm{Cov}\{h(X), \phi_{BC}\} = \kappa\, s\, \Omega_{hh} b$. Hence

$$\underbrace{J^{\top}\Omega_{hh}J}_{\text{moment variance}} \;\; \underbrace{-\,2J^{\top}\mathrm{Cov}\{h(X), \phi_{BC}\}}_{\text{cross term}} \;=\; (1-2\kappa)\, s^2 b^{\top}\Omega_{hh} b \;=\; (1-2\kappa)\,\mathrm{Var}_T\{\tau(X)\}, \tag{5}$$

so the net omission is $(1-2\kappa)\mathrm{Var}_T\{\tau(X)\}/n_T$. **The sign is fixed by κ analytically and the cancellation at $\kappa = 1/2$ is a property of the design, not something the simulation discovered.** This was pointed out in peer review; an earlier version of this paper claimed the simulation located the cancellation point, and that claim is withdrawn.

Regressing the measured omitted variance on Equation 5 across **all** 216 scenarios with effect modification, not the restricted subset of Section 4.2, gives a slope of 1.003 (SE 0.042, $R^2 = 0.72$), confirming the identity to first order.

The first-order **magnitude** is fixed by the design as well, since $\mathrm{SD}_T(\tau)$, κ and n_T are all inputs. What the simulation contributes is therefore narrower than an earlier version of this paper claimed: the omitted component's *fraction of total variance*, which depends on source information the identity does not fix; its *coverage* consequences; the finite-sample departures from Equation 5; and the behavior of estimators that can only add the positive term. On the negative controls, where the population omission is exactly zero, that finite-sample departure moves the model standard error by a median of 0.26% and at most 1.49%.

κ here imposes exact non-negative proportionality between the source and target modifier coefficient vectors. Modifier sets differing in direction, overlapping only partially, or orthogonal are **not** studied, so κ is not a general alignment parameter and no claim is made about those cases.

3 Methods

The protocol was registered before the run at `studies/MIS-03-target-moment-uncertainty/protocol.md`. Reporting follows ADEMP (8).

Covariates are trivariate normal, unit variances, equicorrelation; source centered at 0 with correlation 0.30, target at d with correlation ρ_T. Outcomes are continuous with additive effects and $N(0, 1)$ errors. The target trial's participants are generated and discarded; the estimator receives n_T, three means, three standard deviations, and $\hat{\theta}_{BC}$ with its standard error.

Effect-modification strength is calibrated per scenario so the treatment effect has a prespecified standard deviation in the *target* population, because the quadratic coefficients are half the linear ones and fixed coefficients would make the overlap factor change effect-modification strength at the same time.

Factor	Levels
Source size n_S	500, 2000
Target size n_T	200, 500, 2000
Overlap d	0, 0.4, 0.8
Effect modification $\mathrm{SD}_T(\tau)$	0, 0.45, 0.90
Alignment κ	0, 0.5, 1
Target correlation ρ_T	0.30, 0.60

252 scenarios, 5000 replicates each. The $\mathrm{SD}_T(\tau)$ levels are labelled by their numerical values rather than as "ordinary" and "strong": no empirical calibration to published MAIC applications was available, and value-laden labels would imply one.

n_S is a factor because the omitted term is $J^\top \Omega J/n_T$ against a retained term of order $1/\mathrm{ESS}_S$, so what governs the ratio is source versus target information rather than n_T alone. A pilot of 120 replicates per configuration over a 16-cell grid crossing $n_S \in \{500, 2000\}$, $n_T \in \{200, 2000\}$, $d \in \{0, 0.8\}$ and two effect-modification strengths, run before the design was fixed, found the omission was 1% to 7% of variance at $n_S = 500$; the 23% figure came from $n_S = 2000$, $n_T = 200$, $d = 0.8$ at the stronger modification. Those numbers justified making n_S a factor and are reported here so the justification is checkable.

Estimand. The target-superpopulation marginal A versus B mean difference, $(1-\kappa)s(0.30d + 0.15d^2)$, exact in closed form and verified against a 3×10^6-draw evaluation in every cell to within 2×10^{-4}. Conditioning on the reported moments is *correct* when the estimand is defined by those moments or they are exact administrative counts; the problem exists only for a superpopulation estimand.

3.1 Methods compared

All four share the same point estimate, so the comparison isolates the interval.

Method	Interval variance	Corrects
target-fixed (status quo)	$V_S + \widehat{\mathrm{Var}}(\hat{\theta}_{BC})$	nothing
normal-recon	$+ J^\top \Omega_{\mathrm{norm}} J/n_T$	positive term only
reported-cov	$+ J^\top \Omega_{hh} J/n_T$	positive term only
joint-score	Equation 4	both terms

Ω_{norm} is reconstructed from the reported means and standard deviations and a borrowed correlation under a multivariate normal model. It is the only one an analyst can compute today.

`normal-recon` and `reported-cov` are partial corrections, and they help over a narrower range than we first stated. They add $M = J^\top \Omega J/n_T$, which is non-negative, and omit the cross term. Writing the correct increment as $(1-2\kappa)M$, the status quo is wrong by $|1-2\kappa|M$ and the partial method is wrong by $2\kappa M$. The partial method is therefore closer to the right variance only when $2\kappa < |1-2\kappa|$, that is when

$$\kappa < 1/4. \tag{6}$$

They tie at $\kappa = 1/4$ and are **worse than doing nothing** for $1/4 < \kappa < 1/2$, a range in which an earlier version of this paper said they helped. Peer review supplied this correction.

The design has $\kappa \in \{0, 0.5, 1\}$ and therefore **no interior values with which to test Equation 6 empirically**. Equation 6 is an analytic result with no empirical support at the boundary it defines, and both reviewers identified that as the largest gap between what this paper recommends and what it tests. Results are reported stratified by κ and no claim is made about intermediate alignment.

`joint-score` receives Ω_T computed inside the simulation and passed **as a matrix only**; it is an enhanced-reporting benchmark, not an assumption that microdata are available.

4 Results

Convergence was at least 99.96% in every one of the 252 scenarios. The largest absolute bias was 0.052 empirical standard errors and coverage and bias-eliminated coverage never differed by more than 0.24 percentage points, so what follows is a property of the interval and not of the point estimate.

4.1 The registered result: uninformative

The protocol declares the run uninformative if the reference method or any negative control leaves 0.93 to 0.97. Across the grid that condition is met, and **the registered conclusion is uninformative.**

The cause is a second failure worth reporting on its own account. At $d = 0.8$ the effective sample size falls to a median of about 148 of 500 and the Wald sandwich undercovers for *every* method, including scenarios with $\mathrm{SD}_T(\tau) = 0$ (Figure 1). There the reported moments carry no population information about the transported effect, J is zero in expectation, and all four intervals should very nearly coincide. They do coincide, and at poor overlap they are wrong together by up to seven percentage points.

This bears directly on the framing in section 1. (3) found conventional estimators anticonservative under poor and moderate overlap; the conventional sandwich used here fails in exactly that region. Source weight-estimation variance is therefore not "settled" in the sense of a procedure that works everywhere, and any absolute-coverage claim outside the region where the controls hold is unsupported.

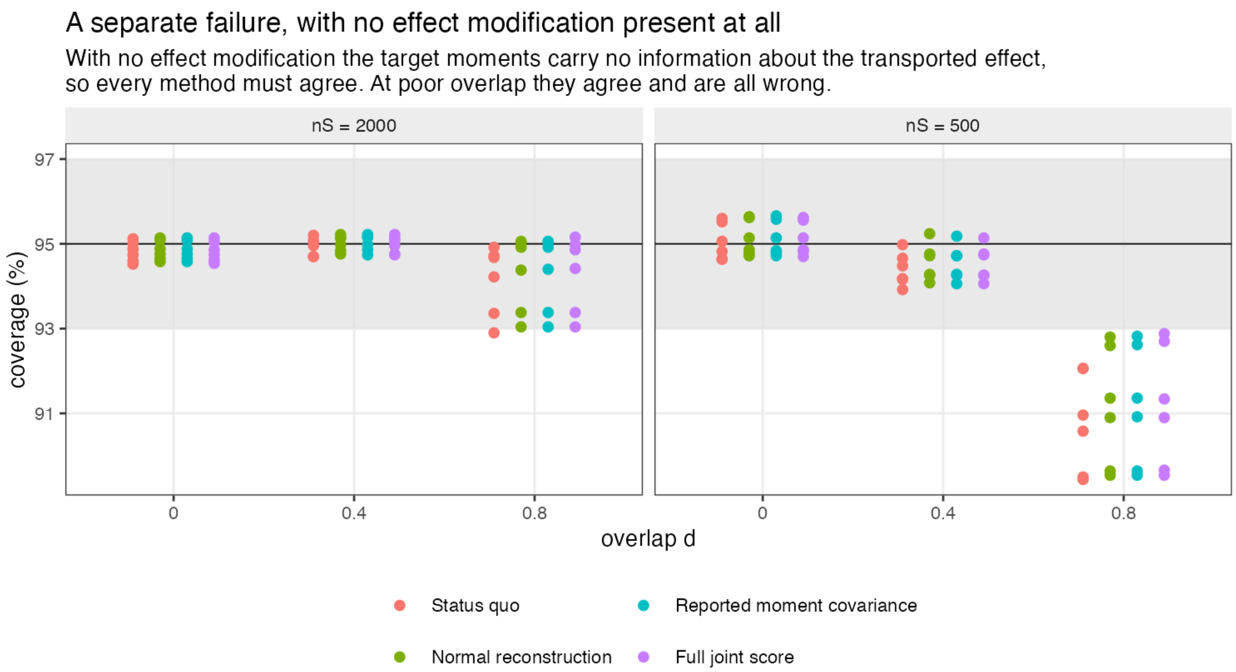

Figure 1: With no effect modification the reported moments carry no population information about the transported effect, so all four methods should nearly coincide. They do, and at poor overlap they are all wrong together.

4.2 The exploratory restriction

Applying the same criterion per scenario leaves **196 of 252** scenarios in which the reference and its matched negative control are both nominal.

This selects on an observed outcome. The selecting quantity is the reference method's coverage, which shares replicates with every other method through the common point estimate and source sandwich, so the reference method's coverage range within the restricted set is partly guaranteed by the selection. Everything in this section is exploratory, is **hypothesis-generating only**, and should not be cited as evidence for or against any method's coverage properties. A confirmatory answer needs a fresh run under a new registration, with either an ex ante design criterion such as expected effective sample size or an overlap restriction fixed in advance, or a source variance procedure validated to pass the controls throughout. We did not re-restrict by an ex ante criterion and report whether the conclusions change; that is part of the same required rerun.

Table 3: Variance omitted by the status quo, stratified by alignment. Defined as $100(\overline{\mathrm{SE}}_{\mathrm{joint}}/\overline{\mathrm{SE}}_{\mathrm{fixed}})^2 - 1$, where each $\overline{\mathrm{SE}}$ is that method's mean model standard error within the scenario. Positive means the reported interval is too narrow. Exploratory.

Alignment	Minimum	Median	Maximum	Sign predicted by identity
0.0	0.9	4.1	14.9	positive
0.5	-0.0	0.7	2.5	zero
1.0	-9.5	-2.3	0.0	negative

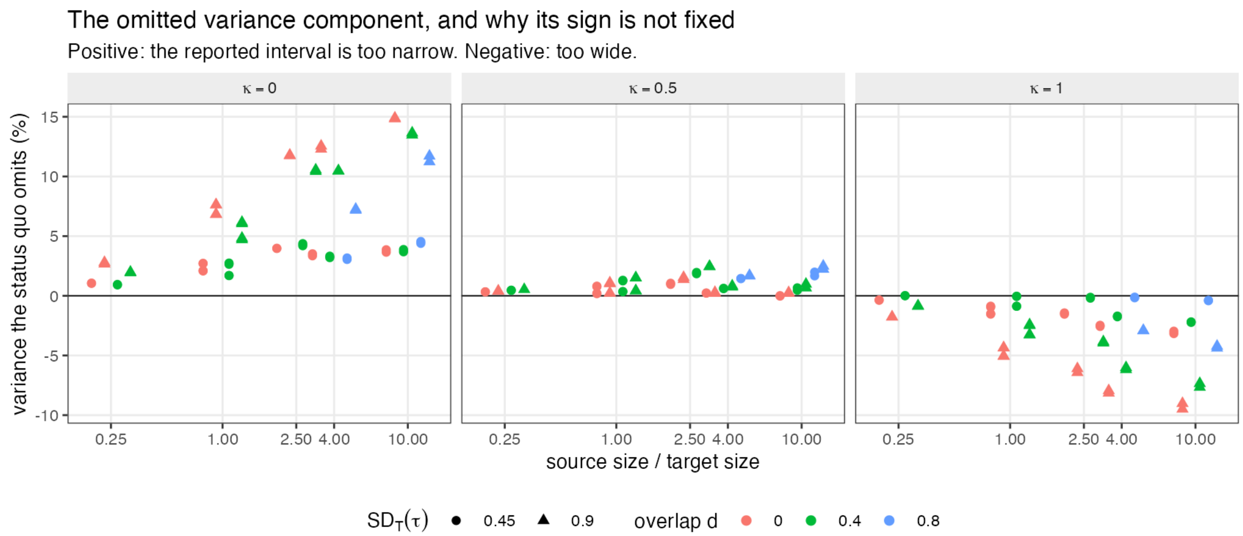

Figure 2: The omitted variance component. Magnitude grows with the ratio of source to target information; the sign follows Equation 5.

Magnitude tracks the ratio of source to target information, as Equation 3 predicts: the omitted term scales as $1/n_T$ and the retained term as $1/\mathrm{ESS}_S$, so a large source trial matched to a small target publication is the worst case, not a small target trial as such.

Table 4: Coverage of nominal 95% intervals across the restricted scenarios with effect modification present, stratified by alignment. These are the 167 of the 196 restricted scenarios that have effect modification; the other 29 are the negative controls. Monte Carlo standard error is 0.31 percentage points at nominal coverage. Exploratory.

Method	kappa=0	kappa=0.5	kappa=1	Outside 93-97%
Status quo	92.1% to 95.0%	92.8% to 95.7%	93.5% to 96.2%	8 of 167
Normal reconstruction (partial)	93.2% to 95.6%	93.5% to 96.5%	93.7% to 97.0%	0 of 167
Reported moment covariance (partial)	93.2% to 95.4%	93.5% to 96.3%	93.7% to 97.0%	0 of 167
Full joint score	93.2% to 95.3%	93.2% to 95.8%	93.4% to 95.5%	0 of 167

The minimum paired differences in Table 5 are within about one and a half Monte Carlo standard errors of zero and carry no directional interpretation; only the maxima are large relative to Monte Carlo error.

The status quo is the only method to leave the band, and it leaves in both directions. The two partial corrections behave as Equation 5 requires: they help at $\kappa = 0$, are mildly conservative at $\kappa = 0.5$ where the net omission is zero, and are more conservative at $\kappa = 1$ where the interval was already too wide.

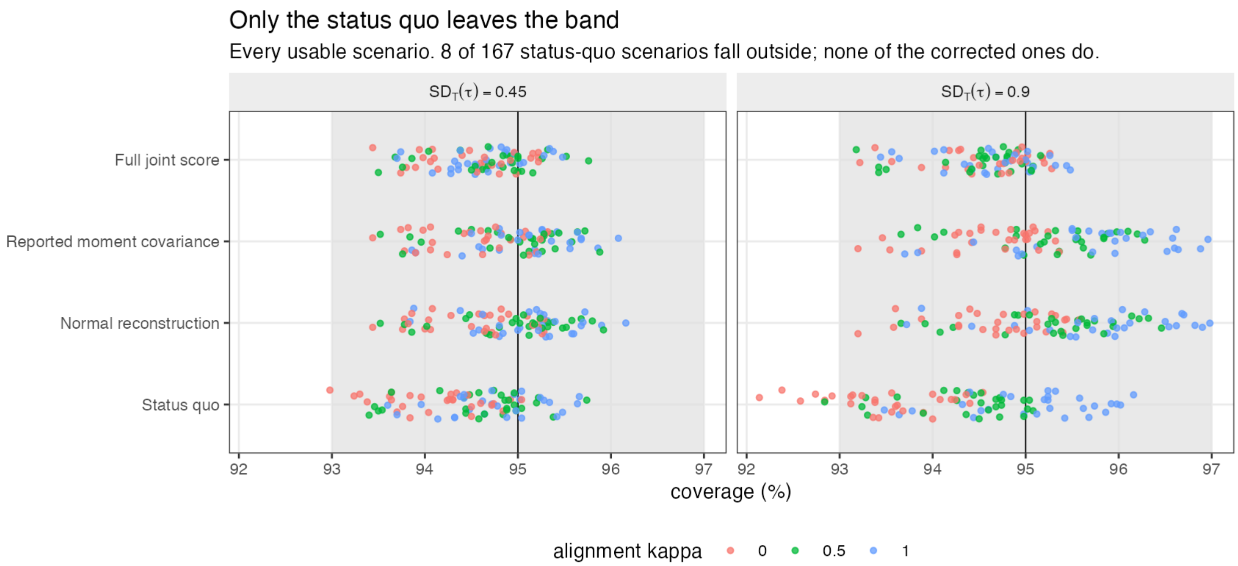

Figure 3: Only the status quo leaves the band.

Figure 4: Status-quo coverage error against the ratio of source to target information.

Table 5: Paired coverage differences against the full joint score, in percentage points, with the discordance counts that determine their Monte Carlo error. A paired difference is a McNemar contrast on shared replicates, so its Monte Carlo error is governed by the discordant pairs and not by the marginal coverages. Exploratory.

Method	Median	Minimum	Maximum	Mean discordant replicates	Mean MCSE (pp)
Status quo	-0.12	-2.12	1.04	25	0.09
Normal reconstruction (partial)	0.32	-0.12	2.00	27	0.09
Reported moment covariance (partial)	0.28	-0.12	1.92	25	0.09

The single scenario that triggered the prespecified "material at ordinary strength" category was $n_S = 2000$, $n_T = 500$, $d = 0.8$, $\mathrm{SD}_T(\tau) = 0.45$, $\kappa = 0$, $\rho_T = 0.30$, with coverage 0.930 against a Monte Carlo standard error of 0.36 percentage points. It sits at the poor-overlap boundary where the sandwich is already marginal, and no weight is placed on it.

4.3 What a publication would have to report

The normal reconstruction needs nothing beyond what is published plus a correlation matrix borrowed from the source. Within the restricted scenarios it kept coverage inside 93% to 97% everywhere, at a median interval-width cost of 2.24% and a maximum of 8.70%, and its paired difference from being handed the target's true moment covariance was small relative to the differences from the benchmark (Table 5, Figure 5). Interval-width cost is $100(\overline{w}_{\text{recon}}/\overline{w}_{\text{fixed}} - 1)$ with $\overline{w}$ the scenario-mean interval width. Both this and the omitted-variance measure are ratios of scenario summaries, not means of replicate-level ratios; extrema across scenarios are reported without a multiplicity adjustment and should be read as the range observed rather than as estimates of a worst case.

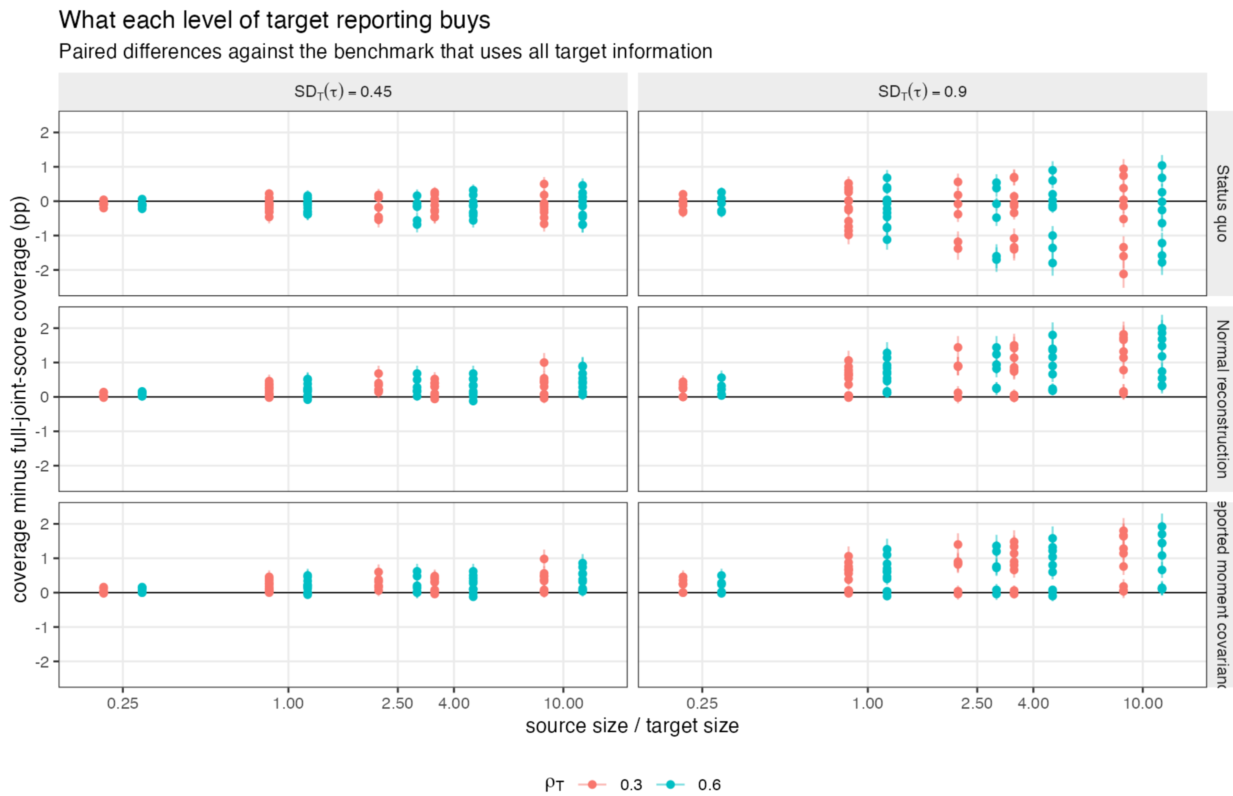

Figure 5: What each level of target reporting buys, as paired differences against the benchmark using all target information.

The recommendation this licenses is narrow. Covariates here are multivariate normal, which is the assumption under which the reconstruction formulas are exactly correct, and the only misspecification examined moves one positive equicorrelation from 0.30 to 0.60. Real baseline tables contain skewed, bounded, categorical, rounded and partially missing variables, and nothing here speaks to those. For approximately normal continuous covariates with modest correlation error, and where κ is believed below about one quarter, adding $J^\top \Omega_{\mathrm{norm}} J/n_T$ costs about 2% of interval width and removes a real omission. Where κ may be large the same addition makes a conservative interval more conservative. Correcting that case needs the covariance between the target's covariate moments and its treatment-effect estimate, which the usual inputs do not supply; we are not aware of a deployable correction, though reported subgroup effects or interaction estimates might in principle carry some of the required information, and we have not investigated that.

An earlier version of this paper recommended the addition generally and concluded that the case for journals publishing moment covariance matrices is weak. Both claims were withdrawn in peer review: the first because the correction is partial, the second because the comparison establishing it was run under conditions favorable to the reconstruction.

5 What this answers, and what it does not

What is established, and by what. Two different things, and they should not be run together. *Analytically*: the omitted component is $(1-2\kappa)\mathrm{Var}_T\{\tau\}/n_T$ to first order, its sign is set by κ, and a correction adding only the positive term helps only for $\kappa < 1/4$. That holds wherever the mechanism's assumptions hold and does not depend on the simulation. *Empirically, and only in the restricted exploratory set*: the fraction of total variance this represents, the resulting coverage, and the finite-sample departures. **The registered coverage question is not confirmatorily answered**, because every usable coverage result comes from a post-run restriction that conditions on an observed outcome. A confirmatory answer requires the fresh registered run described in section 5.2.

Does not answer. Skewed, bounded, categorical, rounded or missing covariates. Binary and time-to-event outcomes, where MAIC is most used and noncollapsibility adds a term this design cannot see. Bias from unmeasured effect modifiers, since modification lies exactly in the matched span. Target summaries drawn from a sample separate from the treatment-effect estimate, or a covariate-adjusted published effect with a different influence function. STC, ML-NMR, ML-UMR and NMI, which have different sensitivities J. And absolute coverage at poor overlap, where the source variance estimator fails first.

Accordingly, and stated precisely rather than as "answers in part": this study contributes an **analytic** result to MIS-03, Equation 5 and Equation 6, which the simulation confirms as theory and which does not depend on the restricted set; and it contributes **exploratory only** evidence on the coverage question MIS-03 actually asks, which is therefore **not** answered confirmatorily. It answers one named component of EST-07, sampling error in reported moments, on the same footing. EST-07's remaining components, model uncertainty in reconstructed correlations, inclusion-criteria ambiguity and secular drift, are not sampling error and are untouched.

Where κ might come from. The practical recommendation turns on κ, which the usual inputs do not identify. Published subgroup effects, reported treatment-by-covariate interactions, or covariate-outcome summaries in the target trial could in principle carry information about how far the target treatment's effect is modified by the same covariates. Establishing what those inputs identify, and how much of κ they pin down, would make the correction deployable in the range where it currently is not. We have not investigated it.

A finding nobody asked for. The poor-overlap sandwich failure in Section 4.1 is larger than the effect this study was built to measure and deserves its own study.

6 Peer review

This manuscript was reviewed in two rounds. Reports, the authors' point-by-point responses and the editorial decision are published in full at `studies/MIS-03-target-moment-uncertainty/review/peer-review.md`. Review changed the paper substantially: the registered result is now reported first and the restricted analysis is labelled exploratory; the sign of the effect is presented as the analytic identity it is rather than as a finding; the two deployable corrections are relabelled partial and the general recommendation to use them is withdrawn; and a factual claim that the protocol had anticipated the sandwich failure was found to be wrong and retracted.

7 Reproducibility

R: R version 4.6.0 (2026-04-24) Platform: aarch64-apple-darwin23 Running under: macOS Tahoe 26.5

7.1 Packages

package	version
base	4.6.0
stats	4.6.0
future	1.70.0
furrr	0.4.0

7.2 Run

- **study**: MIS-03 / EST-07 target-moment uncertainty

- **scenarios**: 252
- **replicates_per_scenario**: 5000
- **total_replicates**: 1260000
- **master_seed**: 20260727

Replicate seeds derive from one master seed through L'Ecuyer-CMRG streams, so a run gives identical results on any number of cores and an interrupted run resumes. Code, protocol, review history and results are at https://github.com/choxos/ITC-open-problems/tree/main/studies/MIS-03-target-moment-uncertainty.

```
Rscript R/03-run.R         # 252 scenarios, resumable
Rscript R/04-analyze.R
Rscript R/05-figures.R
```

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